

7(a)(i)	gain of amplifier is very large	B1
	V^+ is at earth (potential)	B1
	for amplifier not to saturate	M1
	difference between V^- and V^+ must be very small or V^- must be equal to V^+	A1
	or	
	if $V^- \neq V^+$ then feedback voltage	(M1)
	acts to reduce gap until $V^- = V^+$ when stable	(A1)
7(a)(ii)	input impedance is infinite	B1
	(so) current in R_1 = current in R_2	B1
	$(V_{IN} - 0) / R_1 = (0 - V_{OUT}) / R_2$	B1
	(gain =) $V_{OUT} / V_{IN} = -R_2 / R_1$	B1
7(b)	graph: correct inverted shape (straight diagonal line from (0,0) to a negative potential, then a horizontal line, then a straight diagonal line back to the t -axis at the point where $V_{IN} = 0$)	B1
	horizontal line at correct potential of (-)9.0V	B1
	both ends of horizontal line occur at correct times (coinciding with when $V_{IN} = 2.0V$)	B1